Review of Lectures 1–6 and Introduction to Multivariable Integration
Lecture 7 for Multivariable Calculus

Contents

1 Lecture Scope and Goals

This lecture has three purposes:

- review the main concepts and methods from Lectures 1–6,
- collect representative problems with clean solution strategies,
- build a bridge from differential ideas to multivariable integration.

Central Theme

Lectures 1–6 mostly studied **local structure**: vectors, geometry, limits, derivatives, tangent approximations, extrema, and local solvability. Multivariable integration studies the complementary question: how local quantities accumulate to produce a **global total**.

2 Review of Main Concepts and Techniques from Lectures 1–6

2.1 Lecture 1: Vectors, Dot Product, Cross Product, Determinants, Matrices

Main concepts.

- Vectors represent displacement, direction, and magnitude in $\mathbb{R}^2$ and $\mathbb{R}^3$.
- The dot product measures alignment, angle, and projection.
- The cross product in $\mathbb{R}^3$ produces a normal vector and measures area.
- Determinants measure signed area or signed volume scaling.
- Matrices act as linear transformations on vectors.

Techniques to remember.

- To project $\mathbf{u}$ onto $\mathbf{v}$, use
$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}.$$
- To find a plane through three points, form two direction vectors and take a cross product.
- To compute area of a parallelogram, use $\|\mathbf{u} \times \mathbf{v}\|$.
- To compute area of a triangle, divide the parallelogram area by 2.

2.2 Lecture 2: Analytic Geometry in $\mathbb{R}^3$, Quadrics, Parametrizations

Main concepts.

- Surfaces are often recognized by traces and level sets.
- Quadrics are identified by sign patterns and standard forms.
- Parametrizations convert geometric objects into functions of one or two parameters.
- Parametric descriptions are the natural language for curves, surfaces, and later integration.

Techniques to remember.

- To sketch a surface, slice with planes $x = c$, $y = c$, and $z = c$.
- To parametrize an intersection curve, use one equation to choose parameters and the other to determine the remaining variable.
- Always check whether your parametrization covers the whole object or only part of it.

2.3 Lecture 3: Functions of Several Variables, Limits, Continuity, Differentiability

Main concepts.

- A function $f : \mathbb{R}^2 \to \mathbb{R}$ has a domain, graph, and level curves.
- Limits in several variables depend on *all* nearby paths.
- Continuity means the limit equals the function value.
- Differentiability means **best linear approximation**, not just existence of partial derivatives.

Techniques to remember.

- When proving a limit exists, try to dominate the expression by a multiple of $\|(x, y)\|^\alpha$.
- When disproving a limit, look for two paths with different limiting values.
- Testing only lines is not sufficient; curves such as $y = x^2$ may reveal the truth.
- Partial derivatives may exist even when the function is not differentiable.

2.4 Lecture 4: Directional Derivatives, Gradient, Chain Rule, Jacobian

Main concepts.

- The directional derivative measures rate of change in a chosen direction.
- The gradient points in the direction of steepest increase.
- The tangent plane comes from the linear approximation.
- The chain rule and Jacobian organize derivatives of composite maps.
- Lagrange multipliers introduce constrained optimization through gradients.

Techniques to remember.

- Normalize the direction vector before using

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}.$$

- For $z = f(x, y)$, the tangent plane at $(a, b, f(a, b))$ is

$$z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

- For composite dependence, draw the variable-dependence diagram before differentiating.
- For a vector map F, use the Jacobian matrix as the linear part of the approximation.

2.5 Lecture 5: Local and Absolute Extrema

Main concepts.

- Interior local extrema require $\nabla f = 0$ or non-differentiability.
- The Hessian test classifies many interior critical points.
- On a closed and bounded region, absolute extrema exist for continuous f.
- Boundary analysis is essential in global optimization.
- Lagrange multipliers handle smooth equality constraints.

Techniques to remember.

- For absolute extrema on a closed region: check interior, boundary, and corner/end points.
- For the Hessian test in $\mathbb{R}^2$, compute

$$D = f_{xx} f_{yy} - (f_{xy})^2.$$

- For a constraint $g(x, y) = c$, solve

$$\nabla f = \lambda \nabla g$$

together with the constraint equation.

2.6 Lecture 6: Taylor Expansion, Inverse Function Theorem, Implicit Function Theorem

Main concepts.

- Taylor expansion refines linear approximation to higher order.
- A nonzero Jacobian determinant signals local invertibility.
- A nonzero partial derivative such as $F_y \neq 0$ allows local solution of $F(x, y) = 0$ for one variable.
- Higher-order terms help when the Hessian test is inconclusive.

Techniques to remember.

- Keep only terms up to the requested order in a Taylor polynomial.
- For implicit differentiation,

$$F_x + F_y\, y'(x) = 0 \quad \Longrightarrow \quad y'(x) = -\frac{F_x}{F_y}.$$

- The inverse and implicit function theorems are local theorems, not global ones.
- When the quadratic test fails, inspect higher-order Taylor terms or strategic paths.

Master Review Checklist

When you face a new problem, first ask:

1. Is this a geometry problem, a limit problem, a derivative problem, or an optimization problem?
2. What is the right local object: vector, gradient, Jacobian, Hessian, or Taylor polynomial?
3. Is the issue local or global?
4. Do I need a formula, a geometric interpretation, or both?
5. Am I checking every candidate point, every boundary piece, or every relevant path?

3 Typical Problems, Solutions, and Pitfalls

3.1 Problem 1: Plane Through Three Points and Area of a Triangle

Let

$$A = (1, 0, 2), \qquad B = (3, -1, 1), \qquad C = (0, 2, 4).$$

Find:

1. the equation of the plane through A, B, C,
2. the area of triangle ABC.

Solution. Form two direction vectors:

$$\overrightarrow{AB} = B - A = (2, -1, -1), \qquad \overrightarrow{AC} = C - A = (-1, 2, 2).$$

A normal vector is

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & -1 \\ -1 & 2 & 2 \end{vmatrix} = (0, -3, 3).$$

We may simplify this to

$$\mathbf{n} = (0, -1, 1).$$

Using point $A = (1, 0, 2)$, the plane equation is

$$\mathbf{n} \cdot (x - 1, y - 0, z - 2) = 0,$$

so

$$0(x - 1) - y + (z - 2) = 0.$$

Therefore the plane is

$$z - y = 2.$$

For the triangle area,

$$\text{Area of parallelogram} = \left\| \overrightarrow{AB} \times \overrightarrow{AC} \right\| = \|(0, -3, 3)\| = 3\sqrt{2}.$$

Hence

$$\text{Area of triangle } ABC = \frac{1}{2} \cdot 3\sqrt{2} = \frac{3\sqrt{2}}{2}.$$

Pitfall Alert

- Reversing the order of the cross product changes the sign of the normal, but both normals define the same plane.
- Do not forget the factor $\frac{1}{2}$ for the area of a triangle.
- A normal vector alone is not enough; you still need a point on the plane.

3.2 Problem 2: Parametrizing an Intersection Curve

Parametrize the curve of intersection of

$$x^2 + y^2 = 4 \qquad \text{and} \qquad z = x - 1.$$

Then find a tangent vector at the point $(0, 2, -1)$.

Solution. The cylinder suggests circular coordinates:

$$x = 2\cos t, \qquad y = 2\sin t, \qquad 0 \le t < 2\pi.$$

Substitute into the plane equation:

$$z = x - 1 = 2\cos t - 1.$$

So a parametrization is

$$\mathbf{r}(t) = \big(2\cos t,\ 2\sin t,\ 2\cos t - 1\big).$$

To locate $(0, 2, -1)$, we need

$$2\cos t = 0, \qquad 2\sin t = 2.$$

This gives

$$t = \frac{\pi}{2}.$$

Differentiate:

$$\mathbf{r}'(t) = \big(-2\sin t,\ 2\cos t,\ -2\sin t\big).$$

At $t = \frac{\pi}{2}$,

$$\mathbf{r}'\left(\frac{\pi}{2}\right) = (-2, 0, -2).$$

Any nonzero scalar multiple of this vector is also a tangent vector.

Pitfall Alert

- A correct parametrization must satisfy *both* equations.
- If you identify the point incorrectly, the tangent vector will be computed at the wrong parameter value.
- A tangent vector is $\mathbf{r}'(t)$, not $\mathbf{r}(t)$ itself.

3.3 Problem 3: A Multivariable Limit Where Testing Lines Is Not Enough

Determine whether

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2}$$

exists.

Solution. First test a simple path:

$$y = 0 \quad \Longrightarrow \quad \frac{x^2 y}{x^4 + y^2} = 0.$$

So along $y = 0$, the expression tends to 0.

Now test the curve

$$y = x^2.$$

Then

$$\frac{x^2 y}{x^4 + y^2} = \frac{x^2(x^2)}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2} \quad (x \neq 0).$$

So along $y = x^2$, the expression tends to $\frac{1}{2}$.

Because two paths give different limiting values, the limit does **not** exist.

Pitfall Alert

- Checking only lines such as $y = mx$ is not enough in two variables.
- To prove a limit exists, path testing can never be sufficient by itself.
- To prove a limit does not exist, one counterexample path is enough.

3.4 Problem 4: Directional Derivative and Tangent Plane

Let

$$f(x, y) = x^2 + xy + y^2.$$

At the point $(1, -1)$:

1. compute the directional derivative in the direction from $(1, -1)$ to $(2, 1)$,
2. find the tangent plane to the surface $z = f(x, y)$.

Solution. First compute the gradient:

$$f_x = 2x + y, \qquad f_y = x + 2y.$$

So at $(1, -1)$,

$$\nabla f(1, -1) = \big(2(1) + (-1),\ 1 + 2(-1)\big) = (1, -1).$$

The direction vector from $(1, -1)$ to $(2, 1)$ is

$$\mathbf{v} = (1, 2).$$

Its length is

$$\|\mathbf{v}\| = \sqrt{1^2 + 2^2} = \sqrt{5},$$

so the unit vector is

$$\mathbf{u} = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right).$$

Therefore

$$D_{\mathbf{u}} f(1, -1) = \nabla f(1, -1) \cdot \mathbf{u} = (1, -1) \cdot \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right) = -\frac{1}{\sqrt{5}}.$$

Next,

$$f(1, -1) = 1 - 1 + 1 = 1.$$

So the tangent plane is

$$z - 1 = f_x(1, -1)(x - 1) + f_y(1, -1)(y + 1).$$

That is,

$$z - 1 = (x - 1) - (y + 1),$$

or equivalently

$$z = x - y - 1.$$

Pitfall Alert

- Directional derivative formulas require a *unit* direction vector.
- The gradient and the directional derivative are different objects: one is a vector, the other is a scalar.
- In the tangent-plane formula, use the base point (a, b) consistently.

3.5 Problem 5: Absolute Extrema on a Closed Region

Find the absolute maximum and minimum of

$$f(x, y) = x^2 + y^2 - 2x - 4y$$

on the triangle with vertices $(0, 0)$, $(4, 0)$, and $(0, 3)$.

Solution. Since the triangle is closed and bounded and f is continuous, absolute extrema exist.

Step 1: interior critical point.

$$f_x = 2x - 2, \qquad f_y = 2y - 4.$$

Set these equal to zero:

$$2x - 2 = 0, \qquad 2y - 4 = 0 \implies (x, y) = (1, 2).$$

This point lies inside the triangle, and

$$f(1, 2) = 1 + 4 - 2 - 8 = -5.$$

Step 2: boundary edge $y = 0$.

$$f(x, 0) = x^2 - 2x, \qquad 0 \le x \le 4.$$

This has critical point $x = 1$, giving value

$$f(1, 0) = -1.$$

At endpoints:

$$f(0, 0) = 0, \qquad f(4, 0) = 8.$$

Step 3: boundary edge $x = 0$.

$$f(0, y) = y^2 - 4y, \qquad 0 \le y \le 3.$$

This has critical point $y = 2$, giving

$$f(0, 2) = -4.$$

At endpoints:

$$f(0, 0) = 0, \qquad f(0, 3) = -3.$$

Step 4: boundary edge from $(4,0)$ to $(0,3)$. The line is

$$y = 3 - \frac{3}{4}x, \qquad 0 \le x \le 4.$$

Substitute into f:

$$f\left(x, 3 - \frac{3}{4}x\right) = x^2 + \left(3 - \frac{3}{4}x\right)^2 - 2x - 4\left(3 - \frac{3}{4}x\right).$$

Simplifying,

$$f\left(x, 3 - \frac{3}{4}x\right) = \frac{25}{16}x^2 - \frac{7}{2}x - 3.$$

Differentiate:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{25}{16}x^2 - \frac{7}{2}x - 3\right) = \frac{25}{8}x - \frac{7}{2}.$$

So the edge critical point is

$$x = \frac{28}{25}, \qquad y = 3 - \frac{3}{4}\cdot\frac{28}{25} = \frac{54}{25}.$$

The corresponding value is

$$f\left(\frac{28}{25}, \frac{54}{25}\right) = -\frac{124}{25}.$$

Final comparison. The candidate values are

$$-5, \quad -1, \quad 8, \quad -4, \quad -3, \quad -\frac{124}{25}.$$

Hence:

- the absolute minimum is
$$f(1,2) = -5,$$
- the absolute maximum is
$$f(4,0) = 8.$$

Pitfall Alert

- For absolute extrema on a closed region, interior points alone are not enough.
- Each boundary edge becomes a one-variable optimization problem.
- The final answer must include both the *point* and the *function value*.

3.6 Problem 6: Implicit Differentiation and Local Solvability

Consider

$$F(x,y) = x^2 + xy + y^2 - 3.$$

Show that near $(1,1)$ the equation

$$F(x,y) = 0$$

defines y as a differentiable function of x, and compute $y'(1)$.

Solution. First check that $(1,1)$ lies on the curve:

$$F(1,1) = 1+1+1-3 = 0.$$

Now compute

$$F_y(x,y) = x+2y.$$

At $(1,1)$,

$$F_y(1,1) = 1+2 = 3 \neq 0.$$

Therefore, by the implicit function theorem, near $(1,1)$ the equation $F(x,y)=0$ defines y as a differentiable function of x.

Differentiate

$$x^2+xy+y^2-3=0$$

with respect to x:

$$2x+y+xy'+2yy' = 0.$$

Group the y' terms:

$$(x+2y)y' = -(2x+y).$$

Hence

$$y' = -\frac{2x+y}{x+2y}.$$

At $(1,1)$,

$$y'(1) = -\frac{2(1)+1}{1+2(1)} = -\frac{3}{3} = -1.$$

Pitfall Alert

- The implicit function theorem is local: it guarantees solvability only near the point where the derivative condition holds.
- Check both conditions: $F(a,b)=0$ and the relevant partial derivative is nonzero.
- When differentiating xy, use the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}(xy) = y + xy'.$$

4 Comprehensive Problems: Constructive and Challenging

The following problems are less routine than the earlier example bank. Each one is designed to make students combine ideas from several lectures instead of applying a single formula mechanically.

4.1 Comprehensive Problem 1: Construct a Counterexample to "Directional Derivatives Imply Differentiability"

Define

$$f(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y)\neq(0,0), \\ 0, & (x,y)=(0,0). \end{cases}$$

Study the behavior of f at the origin.

1. Show that all directional derivatives of f at $(0,0)$ exist.
2. Compute $f_x(0,0)$ and $f_y(0,0)$.
3. Show that f is not differentiable at $(0,0)$.
4. Explain clearly why this does not contradict the formulas from the lecture on directional derivatives.

Hint. For a direction $\mathbf{u} = (a, b)$, compute

$$\lim_{t\to 0} \frac{f(ta, tb) - f(0,0)}{t}.$$

If f were differentiable, then every directional derivative would have to agree with

$$\nabla f(0,0) \cdot \mathbf{u}.$$

Compare this prediction with the actual formula.

Solution sketch. For a nonzero direction $\mathbf{u} = (a, b)$,

$$\frac{f(ta, tb) - f(0,0)}{t} = \frac{1}{t} \cdot \frac{t^3 a^3}{t^2(a^2 + b^2)} = \frac{a^3}{a^2 + b^2}.$$

So every directional derivative exists. In particular, if $\mathbf{u}$ is a unit vector, then

$$D_{\mathbf{u}} f(0,0) = a^3.$$

For the partial derivatives,

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - 0}{h} = \lim_{h\to 0} \frac{h^3/h^2}{h} = 1,$$

while

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,h) - 0}{h} = 0.$$

Thus

$$\nabla f(0,0) = (1, 0).$$

If f were differentiable at the origin, its linear part would be

$$L(x, y) = x,$$

so we would have

$$\frac{f(x,y) - x}{\sqrt{x^2 + y^2}} \to 0.$$

But along the path $y = x$,

$$f(x, x) = \frac{x^3}{2x^2} = \frac{x}{2},$$

hence

$$\frac{f(x,x) - x}{\sqrt{x^2 + x^2}} = \frac{-x/2}{\sqrt{2}\,|x|} \to -\frac{1}{2\sqrt{2}},$$

which is not 0. Therefore f is not differentiable at $(0,0)$.

There is no contradiction: the identity

$$D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$$

holds when f is differentiable. Existence of directional derivatives alone is not enough.

4.2 Comprehensive Problem 2: Parametrization, Tangent Vector, and Extrema on a Space Curve

Consider the curve C given by the intersection of the cylinder

$$x^2 + y^2 = 4$$

and the paraboloid

$$z = x^2 + y^2 - 2x.$$

1. Find a parametrization of C.
2. Find a tangent vector at the point $(0, 2, 4)$.
3. Determine the highest and lowest points on the curve.
4. Explain whether your answer is easier to obtain from the parametrization or directly from the defining equations.

Hint. On the cylinder,

$$x = 2\cos t, \qquad y = 2\sin t.$$

Then substitute into the formula for z. Once the curve is parametrized, the height becomes a one-variable function.

Solution sketch. On the cylinder, use

$$x = 2\cos t, \qquad y = 2\sin t.$$

Then

$$z = x^2 + y^2 - 2x = 4 - 4\cos t.$$

So a parametrization is

$$\mathbf{r}(t) = \big(2\cos t,\ 2\sin t,\ 4 - 4\cos t\big).$$

The point $(0, 2, 4)$ corresponds to

$$t = \frac{\pi}{2}.$$

Differentiate:

$$\mathbf{r}\,'(t) = \big(-2\sin t,\ 2\cos t,\ 4\sin t\big).$$

Thus a tangent vector at $(0, 2, 4)$ is

$$\mathbf{r}\,'\Big(\frac{\pi}{2}\Big) = (-2, 0, 4).$$

The height along the curve is

$$z(t) = 4 - 4\cos t.$$

This is smallest when $\cos t = 1$, i.e. at

$$(2, 0, 0),$$

and largest when $\cos t = -1$, i.e. at

$$(-2, 0, 8).$$

So the lowest point on the curve is $(2, 0, 0)$ and the highest point is $(-2, 0, 8)$.

This problem is easier from the parametrization, because it converts a space-curve problem into a one-variable problem in t.

4.3 Comprehensive Problem 3: Local Quadratic Model of an Implicit Curve

Let

$$F(x, y) = x^2 + xy + y^2 - 3.$$

Near the point $(1, 1)$, the equation

$$F(x, y) = 0$$

defines y as a function of x.

1. Verify that the implicit function theorem applies at $(1, 1)$.
2. Compute $y'(1)$.
3. Differentiate once more to compute $y''(1)$.
4. Use this information to write the second-order Taylor approximation of the curve near $(1, 1)$ in the form
$$y(x) \approx 1 + a(x - 1) + b(x - 1)^2.$$
5. Use your approximation to describe the local shape of the curve near $(1, 1)$.

Hint. Start from

$$F(x, y(x)) = 0.$$

Differentiate once to get y', then differentiate again carefully. This problem combines the implicit function theorem with Taylor expansion.

Solution sketch. We have

$$F(1, 1) = 1 + 1 + 1 - 3 = 0$$

and

$$F_y(x, y) = x + 2y, \qquad F_y(1, 1) = 3 \neq 0.$$

So the implicit function theorem applies near $(1, 1)$.

Differentiate

$$x^2 + xy + y^2 - 3 = 0$$

once:

$$2x + y + xy' + 2yy' = 0.$$

Hence

$$y' = -\frac{2x + y}{x + 2y}.$$

At $(1, 1)$,

$$y'(1) = -1.$$

Differentiate the first-derivative equation again:

$$2 + y' + y' + xy'' + 2(y')^2 + 2yy'' = 0.$$

So

$$(x + 2y)y'' = -2 - 2y' - 2(y')^2.$$

At $(1,1)$ with $y'(1) = -1$,

$$3y''(1) = -2 + 2 - 2 = -2,$$

therefore

$$y''(1) = -\frac{2}{3}.$$

The second-order Taylor approximation of $y(x)$ at $x = 1$ is

$$y(x) \approx y(1) + y'(1)(x-1) + \frac{y''(1)}{2}(x-1)^2,$$

so

$$y(x) \approx 1 - (x-1) - \frac{1}{3}(x-1)^2.$$

Thus near $(1,1)$ the curve is decreasing and concave downward; in particular, it lies below its tangent line to second order.

4.4 Comprehensive Problem 4: Linear Maps, Determinants, and Ellipse Geometry

Consider the ellipse

$$E = \left\{ (x,y) \in \mathbb{R}^2 : \frac{x^2}{9} + \frac{y^2}{4} \le 1 \right\}.$$

1. Construct a linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ that sends the unit disk

   $$u^2 + v^2 \le 1$$

   onto E.
2. Compute the matrix of T and its determinant.
3. Explain geometrically why the determinant predicts the area-scaling factor.
4. Use this to predict the area of the ellipse.
5. Compare your answer with the familiar formula πab for an ellipse.

Hint. Stretch by a factor of 3 in one coordinate direction and by a factor of 2 in the other.

Solution sketch. Define

$$T(u,v) = (3u, 2v).$$

If

$$u^2 + v^2 \le 1,$$

then with $x = 3u$ and $y = 2v$ we get

$$\frac{x^2}{9} + \frac{y^2}{4} = u^2 + v^2 \le 1,$$

so T maps the unit disk onto the ellipse E.

The matrix of T is

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix},$$

and

$$\det A = 6.$$

Geometrically, a determinant tells us how a linear map scales signed area. Here the map stretches lengths by 3 in one coordinate direction and by 2 in the other, so areas are multiplied by

$$3 \cdot 2 = 6.$$

Since the unit disk has area π, the ellipse has area

$$6\pi.$$

This matches the familiar ellipse formula

$$\pi ab = \pi(3)(2) = 6\pi.$$

4.5 Comprehensive Problem 5: A Degenerate Critical Point Beyond the Hessian Test

Consider

$$f(x, y) = x^4 + y^4 - x^2y^2.$$

1. Find all critical points of f.
2. Explain why the usual Hessian test at the origin is inconclusive.
3. Classify the critical point at the origin.
4. Decide whether this minimum is strict or non-strict.

Hint. Compute the gradient first. To classify the origin, it is helpful to rewrite the function as a sum of nonnegative terms.

Solution sketch. We compute

$$f_x = 4x^3 - 2xy^2 = 2x(2x^2 - y^2), \qquad f_y = 4y^3 - 2x^2y = 2y(2y^2 - x^2).$$

Critical points satisfy

$$2x(2x^2 - y^2) = 0, \qquad 2y(2y^2 - x^2) = 0.$$

If $x = 0$, then $f_y = 4y^3$, so $y = 0$. If $y = 0$, then $f_x = 4x^3$, so $x = 0$. If $x \neq 0$ and $y \neq 0$, then

$$y^2 = 2x^2, \qquad x^2 = 2y^2,$$

which implies

$$x^2 = 4x^2,$$

hence $x = 0$, a contradiction. Therefore the only critical point is

$$(0, 0).$$

Now compute the Hessian:

$$f_{xx} = 12x^2 - 2y^2, \qquad f_{xy} = -4xy, \qquad f_{yy} = 12y^2 - 2x^2.$$

At the origin,

$$H_f(0,0) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

so the second derivative test is inconclusive.

To classify the point, rewrite

$$f(x,y) = x^4 + y^4 - x^2y^2 = (x^2 - y^2)^2 + x^2y^2.$$

Both terms on the right are nonnegative, so

$$f(x,y) \geq 0$$

for all (x,y). Equality holds only if

$$x^2 - y^2 = 0 \quad \text{and} \quad xy = 0,$$

which forces

$$x = y = 0.$$

Hence $(0,0)$ is a **strict local minimum**; in fact it is a strict global minimum.

4.6 Comprehensive Problem 6: Local Invertibility Versus Global Invertibility

Define

$$F(x,y) = \big(e^x \cos y,\ e^x \sin y\big).$$

1. Compute the Jacobian matrix $J_F(x,y)$.
2. Show that F is locally invertible at $(0,0)$.
3. Compute the Jacobian of the inverse map at the point $F(0,0)$.
4. Explain why F is not globally one-to-one on all of $\mathbb{R}^2$.

Hint. Use the inverse function theorem locally, and then compare $F(x,y)$ with $F(x,y+2\pi)$.

Solution sketch. The Jacobian matrix is

$$J_F(x,y) = \begin{bmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{bmatrix}.$$

Its determinant is

$$\det J_F(x,y) = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x} > 0.$$

In particular,

$$\det J_F(0,0) = 1 \neq 0.$$

Therefore, by the inverse function theorem, F is locally invertible at $(0,0)$.

Also,

$$F(0,0) = (1,0).$$

Since

$$J_F(0,0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

we have
$$J_{F^{-1}}(1,0) = \big(J_F(0,0)\big)^{-1} = I.$$

However, F is not globally one-to-one because
$$F(x, y + 2\pi) = F(x, y)$$
for every (x, y). So different points in $\mathbb{R}^2$ can have the same image. This is a clean example of the difference between local invertibility and global injectivity.

4.7 Comprehensive Problem 7: Tangent Line to a Curve of Intersection

Let the curve C be defined as the intersection of
$$G_1(x, y, z) = x^2 + y^2 - 5 = 0$$
and
$$G_2(x, y, z) = x + z - 3 = 0.$$
Consider the point
$$P = (1, 2, 2).$$

1. Verify that P lies on both surfaces.
2. Find a tangent vector to the curve C at P.
3. Write an equation of the tangent line at P.
4. Find the acute angle between this tangent line and the positive z-axis.

Hint. Each gradient vector is normal to its surface. A tangent vector to the intersection curve must be perpendicular to both normals.

Solution sketch. First,
$$1^2 + 2^2 = 5, \qquad 1 + 2 = 3,$$
so $P = (1, 2, 2)$ lies on both surfaces.

Next,
$$\nabla G_1(x, y, z) = (2x, 2y, 0), \qquad \nabla G_2(x, y, z) = (1, 0, 1).$$
At P,
$$\nabla G_1(P) = (2, 4, 0), \qquad \nabla G_2(P) = (1, 0, 1).$$
A tangent vector is their cross product:
$$\nabla G_1(P) \times \nabla G_2(P) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 4 & 0 \\ 1 & 0 & 1 \end{vmatrix} = (4, -2, -4).$$
So we may use
$$\mathbf{v} = (2, -1, -2)$$
as a tangent vector.

Therefore the tangent line is

$$(x, y, z) = (1, 2, 2) + t(2, -1, -2).$$

In symmetric form, this is

$$\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-2}{-2}.$$

To find the acute angle with the positive z-axis, use the direction vector

$$\mathbf{k} = (0, 0, 1).$$

Then

$$\cos\theta = \frac{|\mathbf{v} \cdot \mathbf{k}|}{\|\mathbf{v}\|\,\|\mathbf{k}\|} = \frac{2}{\sqrt{2^2 + (-1)^2 + (-2)^2}} = \frac{2}{3}.$$

So the acute angle is

$$\theta = \arccos\left(\frac{2}{3}\right).$$

4.8 Comprehensive Problem 8: Proving an Inequality by Lagrange Multipliers

Use Lagrange multipliers to prove that if

$$x, y, z \geq 0 \qquad \text{and} \qquad x + y + z = 3,$$

then

$$xyz \leq 1.$$

Determine when equality holds.

Hint. Maximize

$$f(x, y, z) = xyz$$

subject to the constraint

$$g(x, y, z) = x + y + z = 3.$$

Do not forget to discuss what happens on the boundary of the constraint set.

Solution sketch. Consider the constrained set

$$D = \{(x, y, z) \in \mathbb{R}^3 : x, y, z \geq 0,\ x + y + z = 3\}.$$

This set is closed and bounded, so $f(x, y, z) = xyz$ attains a maximum on D.

If the maximum occurs on the boundary of D, then at least one of x, y, z is 0, so

$$xyz = 0 \leq 1.$$

Thus the only interesting case is an interior point of the constraint set, where

$$x > 0, \qquad y > 0, \qquad z > 0.$$

At such a point we apply Lagrange multipliers. We have

$$\nabla f = (yz, xz, xy), \qquad \nabla g = (1, 1, 1).$$

So

$$\nabla f = \lambda \nabla g$$

means

$$yz = \lambda, \qquad xz = \lambda, \qquad xy = \lambda.$$

Hence

$$yz = xz \quad \Longrightarrow \quad z(y - x) = 0,$$

and since $z > 0$, we get

$$x = y.$$

Similarly,

$$yz = xy \quad \Longrightarrow \quad y(z - x) = 0,$$

so

$$x = z.$$

Therefore

$$x = y = z.$$

Using the constraint,

$$x + y + z = 3 \quad \Longrightarrow \quad 3x = 3,$$

so

$$x = y = z = 1.$$

At this point,

$$xyz = 1.$$

Since boundary points give values at most 0 and the only interior critical point gives value 1, the maximum value is

$$1.$$

Therefore

$$xyz \le 1$$

for all $x, y, z \ge 0$ with $x + y + z = 3$, and equality holds exactly when

$$x = y = z = 1.$$

4.9 Comprehensive Problem 9: Uniqueness of the Multivariable Taylor Expansion

Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is twice differentiable near $(0, 0)$ and satisfies

$$f(x, y) = 1 + 2x - y + ax^2 + bxy + cy^2 + o(x^2 + y^2)$$

as $(x, y) \to (0, 0)$.

Suppose also that from a different calculation you obtained

$$f(x, y) = 1 + 2x - y + (x + y)^2 + 2(x - y)^2 + o(x^2 + y^2).$$

1. Explain why uniqueness of the second-order Taylor expansion forces the two quadratic parts to be equal.

2. Determine a, b, and c.
3. Compute $\nabla f(0,0)$ and the Hessian matrix $H_f(0,0)$.
4. Let
$$g(x,y) = f(x,y) - 1 - 2x + y.$$
Use the Taylor expansion to determine the local behavior of g near $(0,0)$.

Hint. First expand
$$(x+y)^2 + 2(x-y)^2.$$
For the Hessian, remember that the second-order Taylor polynomial has the form
$$f(0,0) + f_x(0,0)x + f_y(0,0)y + \frac{1}{2}\big(f_{xx}(0,0)x^2 + 2f_{xy}(0,0)xy + f_{yy}(0,0)y^2\big).$$

Solution sketch. If two polynomials of degree at most 2 both approximate the same function up to an error
$$o(x^2+y^2),$$
then their difference is also
$$o(x^2+y^2).$$
But a nonzero polynomial of degree at most 2 cannot be $o(x^2+y^2)$ near the origin. Therefore the second-order Taylor polynomial is unique, and the quadratic parts must agree exactly.

Now expand:
$$(x+y)^2 + 2(x-y)^2 = x^2 + 2xy + y^2 + 2x^2 - 4xy + 2y^2 = 3x^2 - 2xy + 3y^2.$$
Hence uniqueness gives
$$a = 3, \qquad b = -2, \qquad c = 3.$$

From the linear part we read off
$$f(0,0) = 1, \qquad f_x(0,0) = 2, \qquad f_y(0,0) = -1,$$
so
$$\nabla f(0,0) = (2,-1).$$

Now compare the quadratic terms with the standard Taylor form:
$$\frac{1}{2}f_{xx}(0,0)x^2 + f_{xy}(0,0)xy + \frac{1}{2}f_{yy}(0,0)y^2 = 3x^2 - 2xy + 3y^2.$$
Therefore
$$f_{xx}(0,0) = 6, \qquad f_{xy}(0,0) = -2, \qquad f_{yy}(0,0) = 6,$$
and hence
$$H_f(0,0) = \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix}.$$

Finally,
$$g(x,y) = f(x,y) - 1 - 2x + y = 3x^2 - 2xy + 3y^2 + o(x^2+y^2).$$
The quadratic form
$$3x^2 - 2xy + 3y^2$$

is positive definite. For example,

$$3x^2 - 2xy + 3y^2 = (x-y)^2 + 2x^2 + 2y^2 > 0$$

whenever $(x, y) \neq (0, 0)$. Therefore

$$g(x, y) > 0$$

for all sufficiently small nonzero (x, y), so g has a strict local minimum at $(0, 0)$.

4.10 Comprehensive Problem 10: A Smooth Function That Is Not Analytic

Define

$$f(x, y) = \begin{cases} e^{-1/(x^2+y^2)}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

1. Show that for every integer $N \geq 1$,

$$f(x, y) = o\big((x^2 + y^2)^N\big) \qquad \text{as } (x, y) \to (0, 0).$$

2. Explain why f is smooth at $(0, 0)$ and why every partial derivative of every order at $(0, 0)$ is equal to 0.
3. Write the Taylor series of f at the origin.
4. Show that this Taylor series does *not* represent f on any neighborhood of $(0, 0)$.
5. Conclude that f is C^∞ but not analytic at the origin.

Hint. Set

$$r^2 = x^2 + y^2.$$

Use the fact that

$$e^{-1/r^2}$$

goes to 0 faster than any power of r as $r \to 0$.

Solution sketch. Let

$$r = \sqrt{x^2 + y^2}.$$

For $(x, y) \neq (0, 0)$ we have

$$f(x, y) = e^{-1/r^2}.$$

Fix $N \geq 1$. Then

$$\frac{f(x, y)}{(x^2 + y^2)^N} = \frac{e^{-1/r^2}}{r^{2N}}.$$

Set

$$t = \frac{1}{r^2}.$$

As $(x, y) \to (0, 0)$, we have $t \to \infty$, and

$$\frac{e^{-1/r^2}}{r^{2N}} = t^N e^{-t} \to 0.$$

Therefore

$$f(x,y) = o\big((x^2+y^2)^N\big)$$

for every N.

Now differentiate away from the origin. Every derivative of f has the form

$$\frac{P(x,y)}{(x^2+y^2)^m}\, e^{-1/(x^2+y^2)}$$

for some polynomial P and some integer $m \geq 0$. The exponential factor decays faster than any power of r, so every such derivative tends to 0 as $(x,y) \to (0,0)$. Hence f extends smoothly to the origin, and all partial derivatives of every order at $(0,0)$ are 0.

Therefore every Taylor coefficient of f at the origin is zero, so the Taylor series is

$$0 + 0x + 0y + 0x^2 + 0xy + 0y^2 + \cdots,$$

that is, the zero series.

However, for every point $(x,y) \neq (0,0)$,

$$f(x,y) = e^{-1/(x^2+y^2)} > 0.$$

So in any neighborhood of the origin, the function is not identically zero. The Taylor series is identically zero, hence it does not agree with f on any neighborhood of $(0,0)$.

Thus f is infinitely differentiable at the origin, but its Taylor series does not recover the function there. So f is **smooth but not analytic** at $(0,0)$.

4.11 Comprehensive Problem 11: Chain Rule, Radial Functions, and the Laplacian

Let $\phi \in C^2([0,\infty))$ and define

$$u(x,y) = \phi(x^2+y^2).$$

1. Use the multivariable chain rule to compute u_x, u_y, and $\Delta u = u_{xx} + u_{yy}$.
2. Apply your formula to
$$u(x,y) = e^{-(x^2+y^2)}.$$
Show that
$$\Delta u = 4(x^2+y^2-1)e^{-(x^2+y^2)}.$$
3. Prove the elementary estimate
$$|\Delta u(x,y)| \leq 4 \qquad \text{for all } (x,y) \in \mathbb{R}^2.$$
4. Explain briefly why this is a good example of how the chain rule organizes derivatives of a composite multivariable function.

Hint. Set

$$s = x^2 + y^2.$$

Then $u(x,y) = \phi(s)$ is a composition of the one-variable function ϕ with the two-variable function $s(x,y) = x^2 + y^2$.

Solution sketch. Let

$$s = x^2 + y^2.$$

Then

$$u(x, y) = \phi(s).$$

By the chain rule,

$$u_x = \phi'(s)\, s_x = \phi'(s) \cdot 2x = 2x\, \phi'(s),$$

and similarly

$$u_y = 2y\, \phi'(s).$$

Differentiate again:

$$u_{xx} = 2\phi'(s) + 2x\frac{\partial}{\partial x}\phi'(s) = 2\phi'(s) + 2x\phi''(s)\, s_x = 2\phi'(s) + 4x^2\phi''(s).$$

Likewise,

$$u_{yy} = 2\phi'(s) + 4y^2\phi''(s).$$

Therefore

$$\Delta u = u_{xx} + u_{yy} = 4\phi'(s) + 4(x^2 + y^2)\phi''(s).$$

Since $s = x^2 + y^2$, this becomes

$$\Delta u = 4\phi'(x^2 + y^2) + 4(x^2 + y^2)\phi''(x^2 + y^2).$$

Now take

$$\phi(s) = e^{-s}.$$

Then

$$\phi'(s) = -e^{-s}, \qquad \phi''(s) = e^{-s}.$$

So

$$\Delta u = 4(-e^{-s}) + 4se^{-s} = 4(s - 1)e^{-s}.$$

Returning to $s = x^2 + y^2$, we get

$$\Delta u = 4(x^2 + y^2 - 1)e^{-(x^2+y^2)}.$$

To prove the estimate, let

$$r^2 = x^2 + y^2.$$

Then

$$|\Delta u(x, y)| = 4|r^2 - 1|e^{-r^2}.$$

Set $q = r^2 \geq 0$. We must show

$$|q - 1|e^{-q} \leq 1.$$

If $0 \leq q \leq 1$, then

$$|q - 1|e^{-q} \leq 1.$$

If $q \geq 1$, define

$$h(q) = (q - 1)e^{-q}.$$

Then

$$h'(q) = e^{-q}(2 - q),$$

so on $[1,\infty)$ the maximum occurs at $q=2$, where

$$h(2)=e^{-2}<1.$$

Hence for all $q\ge 0$,

$$|q-1|e^{-q}\le 1.$$

Therefore

$$|\Delta u(x,y)|\le 4.$$

This example is useful because the Laplacian of a radial function looks complicated if one differentiates blindly, but the chain rule turns it into a clean computation through the intermediate variable

$$s=x^2+y^2.$$

4.12 Comprehensive Problem 12: A Boundary-Extremum Phenomenon for $u_{xx}+u_{yy}=0$

Consider the family of functions

$$u(x,y)=A(x^2-y^2)+2Bxy+Cx+Dy+E$$

on the closed unit disk

$$D=\{(x,y)\in\mathbb{R}^2 : x^2+y^2\le 1\}.$$

1. Verify that

   $$u_{xx}+u_{yy}=0$$

   everywhere in $\mathbb{R}^2$.

2. Find the interior critical points of u.

3. Show that every interior critical point is a saddle point unless u is constant.

4. Use compactness and Lagrange multipliers to show that every nonconstant function in this family attains its absolute maximum and absolute minimum on the boundary circle

   $$x^2+y^2=1.$$

5. Write down the Lagrange multiplier system that determines the boundary extremal points.

Hint. For the interior, use the Hessian matrix. For the boundary, optimize u subject to

$$g(x,y)=x^2+y^2=1.$$

Solution sketch. First compute the second derivatives:

$$u_{xx} = 2A, \qquad u_{yy} = -2A.$$

Therefore

$$u_{xx} + u_{yy} = 2A - 2A = 0.$$

Next compute the gradient:

$$u_x = 2Ax + 2By + C, \qquad u_y = 2Bx - 2Ay + D.$$

So interior critical points satisfy

$$2Ax + 2By + C = 0, \qquad 2Bx - 2Ay + D = 0.$$

Now examine the Hessian:

$$H_u = \begin{bmatrix} 2A & 2B \\ 2B & -2A \end{bmatrix}.$$

Its determinant is

$$\det H_u = (2A)(-2A) - (2B)^2 = -4(A^2 + B^2).$$

If $(A, B) \neq (0, 0)$, then

$$\det H_u < 0,$$

so every interior critical point is a saddle point. Therefore no interior critical point can be a local maximum or local minimum.

If $A = B = 0$, then

$$u(x, y) = Cx + Dy + E$$

is linear. Such a function has an interior critical point only if

$$C = D = 0,$$

in which case u is constant. So for every *nonconstant* function in the family, there is no interior point where an absolute maximum or absolute minimum can occur.

Because D is closed and bounded and u is continuous, absolute extrema do exist on D. Since nonconstant functions in this family have no interior extremum, those absolute extrema must occur on the boundary

$$x^2 + y^2 = 1.$$

To find those boundary points, use Lagrange multipliers with

$$g(x, y) = x^2 + y^2.$$

The system is

$$\nabla u = \lambda \nabla g,$$

that is,

$$\begin{aligned} 2Ax + 2By + C &= 2\lambda x, \\ 2Bx - 2Ay + D &= 2\lambda y, \\ x^2 + y^2 &= 1. \end{aligned}$$

So the absolute maximum and minimum are determined by boundary points satisfying this system.

This calculation shows a striking phenomenon for this whole class of functions: once

$$u_{xx} + u_{yy} = 0$$

and the function is not constant, the extreme values on the disk are forced out to the boundary circle.

4.13 Comprehensive Problem 13: Lagrange Multipliers and the Largest Eigenvalue

Let A be a real symmetric $n \times n$ matrix. Consider the quadratic form

$$q(x) = x^{\mathsf{T}} A x, \qquad x \in \mathbb{R}^n.$$

Show, using Lagrange multipliers, that

$$\max_{\|x\|=1} x^{\mathsf{T}} A x$$

is equal to the largest eigenvalue of A.

Hint. Optimize $q(x)$ subject to the constraint

$$g(x) = x^{\mathsf{T}} x = 1.$$

Use the symmetry of A when computing the gradient of q.

Solution sketch. We maximize

$$q(x) = x^{\mathsf{T}} A x$$

on the unit sphere

$$S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}.$$

Since S^{n-1} is closed and bounded and q is continuous, the maximum exists.

Now compute the gradient. Write

$$q(x) = \sum_{i,j=1}^{n} a_{ij} x_i x_j.$$

Then

$$\frac{\partial q}{\partial x_k} = \sum_{j=1}^{n} a_{kj} x_j + \sum_{i=1}^{n} a_{ik} x_i.$$

Because A is symmetric,

$$a_{ik} = a_{ki},$$

so

$$\frac{\partial q}{\partial x_k} = 2 \sum_{j=1}^{n} a_{kj} x_j.$$

Hence

$$\nabla q(x) = 2Ax.$$

Also,

$$g(x) = x^{\mathsf{T}} x$$

has gradient

$$\nabla g(x) = 2x.$$

Let x_* be a point on the unit sphere where q attains its maximum. By Lagrange multipliers,

$$\nabla q(x_*) = \lambda \nabla g(x_*).$$

Therefore

$$2Ax_* = \lambda(2x_*),$$

so

$$Ax_* = \lambda x_*.$$

Thus x_* is an eigenvector of A, and the corresponding eigenvalue is exactly the Lagrange multiplier λ.

Now evaluate the quadratic form at x_*:

$$q(x_*) = x_*^\mathsf{T} A x_* = x_*^\mathsf{T}(\lambda x_*) = \lambda x_*^\mathsf{T} x_* = \lambda,$$

because $\|x_*\| = 1$. So every constrained critical value of q on the unit sphere is an eigenvalue of A.

Let $\lambda_{\max}$ denote the largest eigenvalue of A. Since A is real symmetric, it has a real eigenvector v with

$$Av = \lambda_{\max} v.$$

Normalize it:

$$u = \frac{v}{\|v\|}.$$

Then $\|u\| = 1$ and

$$q(u) = u^\mathsf{T} A u = \lambda_{\max}.$$

So the maximum value of q on the unit sphere is at least $\lambda_{\max}$.

On the other hand, we already proved that any maximizer must give a value equal to an eigenvalue of A, and no eigenvalue is larger than $\lambda_{\max}$. Therefore

$$\max_{\|x\|=1} x^\mathsf{T} A x = \lambda_{\max}.$$

Remark. The same argument also shows

$$\min_{\|x\|=1} x^\mathsf{T} A x$$

is the smallest eigenvalue of A.

Extension: what if A is not symmetric? Suppose now that A is an arbitrary real $n \times n$ matrix, and consider

$$\max_{\|x\|=1} x^\mathsf{T} A x.$$

Students should *not* expect the answer to be the largest eigenvalue of A, because for a nonsymmetric matrix the eigenvalues may even be complex. The right idea is to reduce the problem to the symmetric part of A.

Write

$$A = S + K, \qquad S = \frac{A + A^\mathsf{T}}{2}, \qquad K = \frac{A - A^\mathsf{T}}{2}.$$

Then S is symmetric and K is skew-symmetric:

$$K^\mathsf{T} = -K.$$

Now for any $x \in \mathbb{R}^n$,

$$x^\mathsf{T} K x = \left(x^\mathsf{T} K x\right)^\mathsf{T} = x^\mathsf{T} K^\mathsf{T} x = -x^\mathsf{T} K x.$$

Since $x^{\mathsf{T}}Kx$ is a real number, this implies

$$x^{\mathsf{T}}Kx = 0.$$

Therefore

$$x^{\mathsf{T}}Ax = x^{\mathsf{T}}Sx$$

for every x. So the maximizing problem becomes

$$\max_{\|x\|=1} x^{\mathsf{T}}Sx,$$

and now S is symmetric, so the previous result applies.

Hence

$$\max_{\|x\|=1} x^{\mathsf{T}}Ax = \lambda_{\max}(S) = \lambda_{\max}\left(\frac{A + A^{\mathsf{T}}}{2}\right).$$

Lagrange-multiplier viewpoint. If

$$q(x) = x^{\mathsf{T}}Ax,$$

then a direct computation gives

$$\nabla q(x) = (A + A^{\mathsf{T}})x = 2Sx.$$

So the constrained critical points satisfy

$$(A + A^{\mathsf{T}})x = 2\lambda x,$$

or equivalently

$$Sx = \lambda x.$$

Thus the critical points are controlled by the symmetric part of A, not by A itself.

Common Pitfall

For a nonsymmetric matrix, it is incorrect to write $\nabla(x^{\mathsf{T}}Ax) = 2Ax$. The correct formula is

$$\nabla(x^{\mathsf{T}}Ax) = (A + A^{\mathsf{T}})x.$$

Teaching Note

These problems work well as board problems, group discussion tasks, or short take-home extensions. They are especially useful because they force students to separate:

- what is true from what is only often true,
- local behavior from global behavior,
- first-order information from higher-order information,
- computational formulas from geometric meaning.

5 How to Differentiate Determinants in Multivariable Calculus

5.1 Determinant as a Multivariable Function

From the multivariable-calculus point of view, the determinant is a function

$$\det : M_n(\mathbb{R}) \to \mathbb{R},$$

where $M_n(\mathbb{R})$ is identified with $\mathbb{R}^{n^2}$ through the entries

$$A = (a_{ij}).$$

Since $\det(A)$ is a polynomial in the entries a_{ij}, it is automatically a smooth function of n^2 real variables.

So we can ask the usual questions:

- what are the partial derivatives $\dfrac{\partial}{\partial a_{ij}} \det(A)$?
- what is the directional derivative of det at A in the direction of a matrix H?
- if $A = A(t)$ depends on a parameter, how do we compute $\dfrac{\mathrm{d}}{\mathrm{d}t} \det(A(t))$?

5.2 Partial Derivative with Respect to One Entry

Fix a matrix

$$A = (a_{ij}).$$

If we vary only one entry a_{ij} and keep all other entries fixed, then the determinant is linear in that entry. Therefore

$$\frac{\partial}{\partial a_{ij}} \det(A) = C_{ij}(A),$$

where $C_{ij}(A)$ is the (i, j)-cofactor:

$$C_{ij}(A) = (-1)^{i+j} M_{ij}(A).$$

Here $M_{ij}(A)$ is the determinant of the matrix obtained by deleting row i and column j.

Why this is true. Expand the determinant along row i:

$$\det(A) = a_{i1}C_{i1}(A) + a_{i2}C_{i2}(A) + \cdots + a_{in}C_{in}(A).$$

When differentiating with respect to a_{ij}, all cofactors in this expansion are constant because they do not depend on the entries of row i. Hence

$$\frac{\partial}{\partial a_{ij}} \det(A) = C_{ij}(A).$$

5.3 Directional Derivative of the Determinant

Let $H = (h_{ij})$ be another $n \times n$ matrix. The directional derivative of det at A in the direction H is

$$D(\det)_A(H) = \sum_{i,j=1}^{n} \frac{\partial \det}{\partial a_{ij}}(A)\, h_{ij}.$$

Using the cofactor formula above,

$$D(\det)_A(H) = \sum_{i,j=1}^{n} C_{ij}(A) h_{ij}.$$

This is the matrix version of the multivariable linear approximation:

$$\det(A + H) = \det(A) + D(\det)_A(H) + o(\|H\|).$$

Equivalent column-wise viewpoint. If the columns of A are $a_1, \ldots, a_n$ and the columns of H are $h_1, \ldots, h_n$, then multilinearity of the determinant gives

$$D(\det)_A(H) = \det(h_1, a_2, \ldots, a_n) \det(a_1, h_2, \ldots, a_n) \cdots \det(a_1, \ldots, a_{n-1}, h_n).$$

So the derivative is obtained by replacing one column at a time by the perturbation column and summing.

5.4 Chain Rule for a Matrix-Valued Path

Now let

$$A = A(t)$$

be a differentiable matrix-valued function. Since det is a smooth function of the n^2 entries, the ordinary chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \det(A(t)) = \sum_{i,j=1}^{n} \frac{\partial \det}{\partial a_{ij}}(A(t))\, a'_{ij}(t).$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t} \det(A(t)) = \sum_{i,j=1}^{n} C_{ij}(A(t))\, a'_{ij}(t).$$

This is already a complete and very usable formula.

5.5 Jacobi's Formula

The previous identity is often written in matrix form as

$$\frac{\mathrm{d}}{\mathrm{d}t} \det(A(t)) = \operatorname{tr}\bigl(\operatorname{adj}(A(t))\, A'(t)\bigr),$$

where $\operatorname{adj}(A)$ is the adjugate matrix, defined as the transpose of the cofactor matrix:

$$\operatorname{adj}(A) = \bigl(C_{ji}(A)\bigr)_{1 \le i,j \le n}.$$

In other words, the (i, j)-entry of $\operatorname{adj}(A)$ is the cofactor $C_{ji}(A)$.

If $A(t)$ is invertible, then

$$\operatorname{adj}(A) = \det(A)\, A^{-1},$$

so Jacobi's formula becomes

$$\frac{\mathrm{d}}{\mathrm{d}t} \det(A(t)) = \det(A(t))\ \operatorname{tr}\bigl(A(t)^{-1} A'(t)\bigr).$$

Important remark. The inverse-based formula requires invertibility, but the cofactor formula

$$\frac{\mathrm{d}}{\mathrm{d}t}\det(A(t)) = \sum_{i,j=1}^{n} C_{ij}(A(t))\, a'_{ij}(t)$$

works even when $A(t)$ is singular.

5.6 Two Short Examples

Example 1: a 2×2 matrix with two variables. Let

$$A(x, y) = \begin{bmatrix} x & 1 \\ y & 2 \end{bmatrix}.$$

Then

$$\det A(x, y) = 2x - y.$$

So directly,

$$\frac{\partial}{\partial x}\det A = 2, \qquad \frac{\partial}{\partial y}\det A = -1.$$

Now check with cofactors:

$$C_{11} = 2, \qquad C_{21} = -1.$$

These are exactly the derivatives with respect to the entries $a_{11} = x$ and $a_{21} = y$.

Example 2: higher-order Taylor expansion at the identity. Let

$$A(t) = I + tB.$$

Then

$$A'(t) = B.$$

At $t = 0$, we have $A(0) = I$, so

$$\det(I) = 1, \qquad I^{-1} = I.$$

Hence Jacobi's formula gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\det(I + tB)\Big|_{t=0} = \operatorname{tr}(B).$$

So the first-order approximation is

$$\det(I + tB) = 1 + t\ \operatorname{tr}(B) + o(t).$$

We can go one step further. Set

$$f(t) = \det(I + tB).$$

Then Jacobi's formula in inverse form gives

$$f'(t) = f(t)\ \operatorname{tr}\big((I + tB)^{-1}B\big).$$

Differentiate again. Using

$$\frac{\mathrm{d}}{\mathrm{d}t}(I + tB)^{-1} = -(I + tB)^{-1}B(I + tB)^{-1},$$

we obtain

$$f''(t) = f'(t)\ \mathrm{tr}\big((I+tB)^{-1}B\big) f(t)\,\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{tr}\big((I+tB)^{-1}B\big)$$

$$= f(t)\Big(\ \mathrm{tr}\big((I+tB)^{-1}B\big)\Big)^2 - f(t)\ \mathrm{tr}\big((I+tB)^{-1}B(I+tB)^{-1}B\big).$$

At $t=0$, this becomes

$$f''(0) = \big(\ \mathrm{tr}(B)\big)^2 - \mathrm{tr}(B^2).$$

Therefore the second-order Taylor expansion at $t=0$ is

$$\det(I+tB) = 1 + t\ \mathrm{tr}(B) + \frac{t^2}{2}\Big(\big(\ \mathrm{tr}(B)\big)^2 - \mathrm{tr}(B^2)\Big) + O(t^3).$$

So near the identity, the determinant is controlled first by the trace, and the second-order correction involves both $(\mathrm{tr}\, B)^2$ and $\mathrm{tr}(B^2)$.

Takeaway

To differentiate a determinant in multivariable calculus, the key facts are:

- det is a smooth function of the matrix entries,
- $\dfrac{\partial}{\partial a_{ij}}\det(A) = C_{ij}(A)$,
- the directional derivative is the cofactor-weighted sum of the perturbation entries,
- for a matrix-valued path $A(t)$, use the chain rule entry by entry,
- when $A(t)$ is invertible, Jacobi's formula gives the compact expression

$$\frac{\mathrm{d}}{\mathrm{d}t}\det(A(t)) = \det(A(t))\ \mathrm{tr}\Big(A(t)^{-1}A'(t)\Big).$$

6 Short Introduction to Multivariable Integration

6.1 Why Integration Is the Next Natural Topic

So far we have focused on **local information**:

- vectors describe direction,
- gradients describe local change,
- tangent planes and Jacobians describe local linear behavior,
- Hessians and Taylor polynomials describe second-order local shape.

Integration asks the opposite question:

How do many small local contributions combine into one total quantity?

This total quantity may represent:

- area,
- signed volume,
- mass with variable density,
- total heat, charge, or probability.

6.2 Double Integrals Over Rectangles

Let

$$R = [a, b] \times [c, d] \subset \mathbb{R}^2$$

and let $f(x, y)$ be a function on R.

We divide the rectangle into many small subrectangles, choose sample points (x_{ij}^*, y_{ij}^*), and form the sum

$$\sum_{i,j} f(x_{ij}^*, y_{ij}^*)\, \Delta A.$$

If these sums approach a limit as the partition is refined, we write

$$\iint_R f(x, y)\, \mathrm{d}A.$$

Interpretation.

- If $f \geq 0$, the double integral gives volume under the surface $z = f(x, y)$ over R.
- If f changes sign, it gives signed volume.
- If f is density, then $\iint_R f\, \mathrm{d}A$ gives total mass.

6.3 Iterated Integrals

On rectangles, the double integral is usually computed by two one-variable integrals:

$$\iint_R f(x, y)\, \mathrm{d}A = \int_a^b \int_c^d f(x, y)\, \mathrm{d}y\, \mathrm{d}x$$

or

$$\iint_R f(x, y)\, \mathrm{d}A = \int_c^d \int_a^b f(x, y)\, \mathrm{d}x\, \mathrm{d}y.$$

The main practical point is:

> A two-dimensional accumulation can often be computed by repeated one-dimensional accumulation.

6.4 A First Example

Compute

$$\iint_{[0,1]\times[0,2]} (x + 2y)\, \mathrm{d}A.$$

Solution. Integrate first with respect to y:

$$\int_0^1 \int_0^2 (x+2y)\ \mathrm{d}y\ \mathrm{d}x = \int_0^1 \left[xy+y^2\right]_{y=0}^{y=2} \mathrm{d}x.$$

So

$$\int_0^1 (2x+4)\ \mathrm{d}x = \left[x^2+4x\right]_0^1 = 1+4 = 5.$$

Therefore

$$\iint_{[0,1]\times[0,2]} (x+2y)\ \mathrm{d}A = 5.$$

6.5 General Regions in the Plane

Later we will integrate over nonrectangular regions such as triangles, disks, and curved domains.

Two common descriptions are:

$$D = \{(x,y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$$

and

$$D = \{(x,y) : c \le y \le d,\ h_1(y) \le x \le h_2(y)\}.$$

Then the double integral becomes

$$\iint_D f(x,y)\ \mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\ \mathrm{d}y\ \mathrm{d}x$$

or

$$\iint_D f(x,y)\ \mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\ \mathrm{d}x\ \mathrm{d}y.$$

6.6 Why Earlier Lectures Matter for Integration

The next chapter will use many earlier ideas again:

- **Lecture 1:** determinants measure area scaling,
- **Lecture 2:** parametrizations describe curves and regions,
- **Lecture 3:** continuity helps guarantee integrability,
- **Lecture 4:** Jacobians organize coordinate changes,
- **Lecture 5:** extrema help estimate integrands,
- **Lecture 6:** local invertibility explains why change of variables works.

For example, in polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

the area element changes by a Jacobian factor:

$$\mathrm{d}A = r\ \mathrm{d}r\ \mathrm{d}\theta.$$

That extra factor r is exactly a determinant phenomenon, so the move from vectors and Jacobians to integration is completely natural.

Preview of What Comes Next

The first serious questions in multivariable integration are:

1. What does $\iint_D f(x,y)\,\mathrm{d}A$ mean geometrically and physically?
2. How do we set up iterated integrals over rectangles and curved regions?
3. When should we switch coordinates, such as from Cartesian to polar?
4. How does the Jacobian determinant correct area and volume under a change of variables?

7 Summary

- Lectures 1–6 developed the language of multivariable calculus: vectors, geometry, limits, derivatives, gradients, Jacobians, extrema, and Taylor expansion.
- Typical problems become manageable when we identify the right tool first: cross product, path test, gradient, Hessian, or implicit differentiation.
- Multivariable integration is the next step because it turns local information into global totals.
- Double integrals begin with small area elements and repeated one-variable integration.
- Determinants, parametrizations, and Jacobians will reappear naturally in change of variables.